PsySuite Manual

version 2.0.6

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Summary

Introduction

Objective

PsySuite is an Android App developed to execute multimodal (audio, visual, tactile) psychophysics tests in commercial smartphones. Its main feature is the ability to deliver stimuli, whose onsets and durations reach a millisecond-level precision and accuracy. In case of multimodal stimulation, cross-modal stimuli' simultaneity can also be guaranteed. The demo version of PsySuite includes three tests and a user interface developed to create a wide range of multi-modal stimuli (allowing a clear-cut evaluation of the App's performance). Tests can also be run by visually impaired people thanks to an audio-tactile interface. All user responses are collected at run-time, saved on device's file system and sent to a dedicated Results Web Manager (***RWM***) to be accessed by registered users.

Disclaimer:

The millisecond-level accuracy of stimuli generation varies across sensory modalities and devices. With unimodal stimuli, PsySuite can guarantee inter-stimuli exact timings and duration in each device without any setup procedure. With bi or tri-modal stimuli, since the stimuli generation process varies across devices, is not guaranteed and must be validated for each smartphone. Presently, it is guaranteed only for these two smartphones:

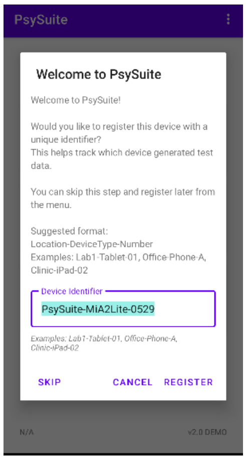
- Samsung A40 (SM-A405FN)
- Xiaomi Mi A2 Lite

Between-modalities stimuli simultaneity can be achieved only following the procedure described in paragraph Device Validation. PsySuite must be used with smartphones or tablets shipped with Android version 8 or above.

Main App features

1. Trimodal Stimulation
    - Acoustic (ring tones or audio files)
    - Tactile (through device vibrators engine, not available on most of the tablets)
    - Visual (graphic symbols visualization)
2. Participants' information collection (label, age, gender, population)
3. Adaptive Design Optimization (ADO) mechanism, to adapt stimuli presentation according to user responses
4. On-device tests' results storing and submission to a dedicated RWM
5. Possibility to divide tests into multiple resumable sessions
6. Audio-tactile user interface for running tests on blind users

First Usage



Users must first grant the following permissions to PsySuite:

- Record audio
- Access external folder

Then, he must define a name for the current device.

This name should be unique and is used to mark each results submission to the Results Web Manager, to distinguish the specific device that collected the results.

Pressing the ⋮ icon in the top-right corner of the screen, users can access the system menu. Pressing "Manage Projects", it is possible to define a set of Projects and associate each experimental session to one of them. In this way, experiments' results submitted to the RWM are categorized under a specific project.

Main window

In the main window, users can select one of the three tests:

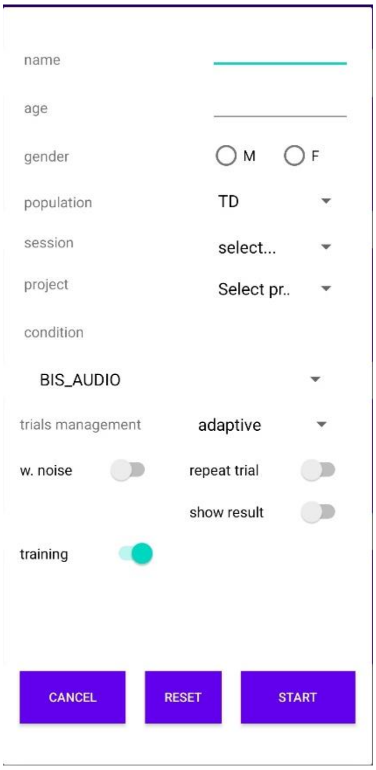
- Reaction Times (RT)
- Temporal Bisection test (BIS)
- Double Flash Illusion Test

Or press the three vertical dots in the top-right corner of the App to:

- open settings windows
- Manage projects

- Send results to RWM
- Device testing
- open user manual.

Subject and experiment information



User must fill in:

- its own demographic info: (name/ID, age, gender, population)
- the current session, in case of longitudinal study
- the reference Project (to classify the results)
- the experimental condition (usually, the sensory modality)

and whether

- use adaptive or constant stimuli
- adding a white noise to each trial (to cancel the vibration-related noise)
- allow user to repeat a trial
- show feedback (success/failure) after each trial
- run a training session

Reaction Times

In Reaction Times test, the App evaluates subjects' responsiveness in pressing a button after having perceived a stimulus in one of three available sensory modalities.

List of subtests or condition

- Audio
- Visual
- Tactile

Temporal Bisection (BIS)

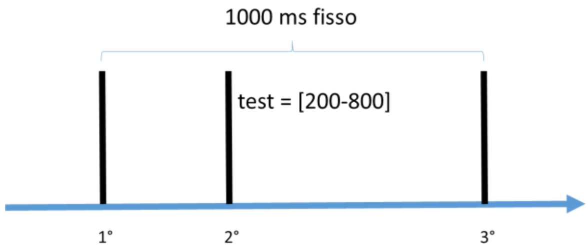
This test can be run in both unimodal and bimodal fashion. The former evaluates the subject's ability to locate stimuli in time. The latter allows us to infer which sensory modality is preferentially located in time. Each of the three unimodal modalities and the three combinations of bimodal modalities are called subtest or condition.

List of subtests or condition

- Audio
- Visual
- Tactile
- Audio-tactile
- Audio-Visual
- Visuo-tactile

Unimodal conditions

The unimodal BIS test consists of three stimuli delivered at three different latencies. The first at 0 ms, the third at 1000 ms, the second at a latency varying from 200 to 800 ms after the first stimulus.



At the end of each trial, participants are asked to indicate whether the second was temporally closer to stimulus 1 or stimulus 3.

Bimodal subtest

These versions investigate a different aspect of temporal discrimination by eventually adding a delay between the two modalities during the second stimulus. In one third this delay is absent, in one third modality A precedes modality B and in the remaining third occurs the contrary. Such delay is under discrimination-threshold, in this way the task evaluates whether subjects are attracted toward one of the two modalities.

Number of trials

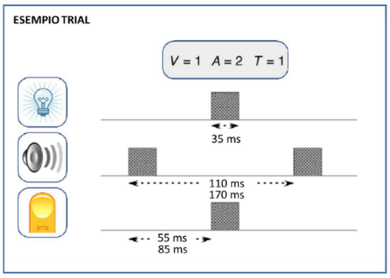
Modality	200	300	400	600	700	800	TOT
unimodal	4	6	6	6	6	4	32
bimodal	4	6	6	6	6	4	32

DFI/TFI test

This test evaluates the user ability to count how many stimuli of each possible modality (acoustic, visual or tactile) are delivered in each trial. The term illusion refers to the fact that specific combinations of stimuli significantly alter the participants' perception of such numbers. Notably, the illusion arises only when stimuli are temporally close to each other (<60 ms). In this test, two different Stimulus-onset asynchronies (SOA) were used, although only the shorter one should be able to induce the illusion.

Trial composition

Each trial is composed by several stimuli interleaved at three fixed temporal onsets. Two different SOA are tested: 55 ms e 85 ms, defining two different sets of combinations (0, 55 and 110 ms, or 0, 85, 170 ms, respectively). Stimuli can be either sounds, flashes, phone vibrations, or can be presented in any possible combination. Stimuli presentation follow these simple rules: if only one stimulus of a specific modality is given in the trial, it occurs at the second onset. If two stimuli of the same type are given, they are delivered at first and third onsets. The number (0,1,2) of stimuli for each modality gives rise to 26 different combinations. Two different SOA are tested: 55 ms e 85 ms, thus defining two different onsets combinations (0, 55 and 110 ms; 0, 85, 170 ms).

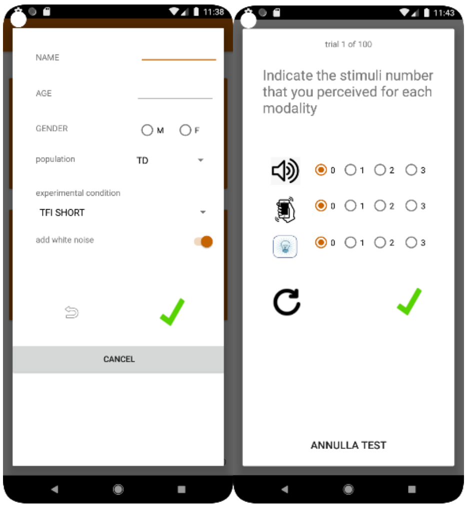


Start test

After pressing the upper button, a new window is opened. There, the user must insert the following participant's information:

- Name (usually an anonymized label or code, without spaces or special characters. Only the underscore "_" character is allowed)
- Age
- Gender
- Population (use TD, typical developing):

A pop-up window informs the user when required information is missing, stopping the procedure until asking to complete it. Since the combination of these information also defines the name of the file containing subject information and test results, if a file with the same name exists, user is informed and asked whether overwriting the existing one or change present user information.



Available subtests

In the experimental condition section, user can select the desired sub-test. Four sub-tests are available:

- TFI: the full trimodal version.
- TFI toddlers: a short trimodal version suitable for young children.
- TFI BI: a version where no more than two modalities are used in each trial.
- DFI AV: an explicit bimodal version using only audio and video stimulation.

A white noise can be constantly added to the test. Used together with headphones, it prevents hearing the sound of the vibration, which may affect the results of audio-tactile tests.
Notably, tablets usually do not have vibration functions, thus the TFI options shall not be available.

Subject response

At the end of each trial, a pop-up window is shown to collect the participant's response. With that, the user can easily indicate the number of perceived stimuli for each sensory modality. Once the user select the desired response, she must press the green tick  to continue with the next trial. Within the current demo version, user can also press the  button to evaluate the same trial again. Testing can be interrupted after any given trial. The App will then ask the user if she wants to keep the results collected so far, or if she wants to delete them. In the former case, user will finally be asked whether sending them via email (not available within the current demo version).

End of a block

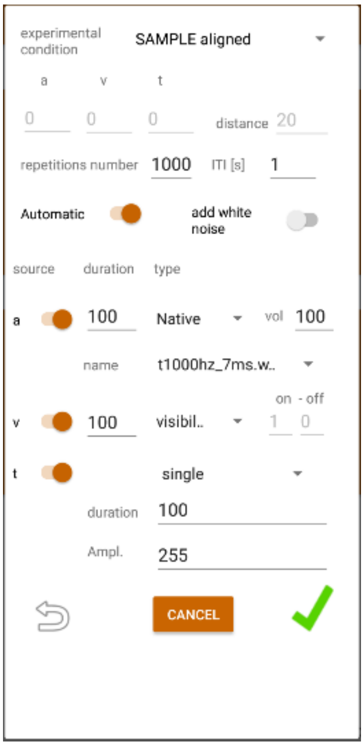
Each test is divided in four blocks. After each block completion, the user is asked whether she wants to continue or to interrupt the session. In the latter case, when the user inserts the same demographic info (as actually expected), the App will recognize the existing previous sessions and will resume the procedure from where it was previously stopped.

End of the test

When all trials are completed (or user ask to stop after a block completion) the result file is stored in the “Download” folder of the device. After that, the file can be freely transferred to be processed with desired software.

Custom Stimuli Creator (CSC)

This user interface has been developed to test and validate PsySuite with the oscilloscope. The user can create a trial composed of either unimodal or multi-modal (audio, visual, tactile) stimuli and deliver it a certain number of times (number of repetition), while choosing an inter-trial interval (ITI) to separate each repetition. Three types of trials can be created:



- Single aligned
- Single shifted
- Pair (aligned)

In “single aligned” type, modalities are delivered simultaneously, while in the “single shifted” version, user can define an offset in milliseconds among each modality. In Pair (aligned), users deliver two uni/multi-stimuli, separated by some milliseconds.

Audio stimuli

There are four modalities to deliver an acoustic stimulus:

- Smartphone tone
- Audio file playback with MediaPlayer API
- Audio file playback with AudioTrack API
- Audio file playback with native (Oboe) API

For the latter three modes, PsySuite comes shipped with 10 different .wav files of different duration and frequency. Developers can of course import and use each type of audio file. To maximize performances (as needed in psychophysical tasks), audio file must not be compressed (a .wav file, encoded in 16-bit PCM mode, is the best option) and sampled at the native device’s sampling frequency (usually 48 KHz). For differences in stimuli stability, please refer to the main manuscript.

Visual stimuli

There are two modalities to deliver an acoustic stimulus:

- Set the visibility of an image resource.
- Show two images' resources (one for ON state, the other for OFF state)

In the present version there are 4 circles of different colors:

- white
- back
- blue
- red

The former is to compare the performance of manipulating the visibility of one resource or alternate two different resources. Developers can add whichever image (uncompressed, if performances care) during task creation.

Tactile stimuli

Users can set up two different types of tactile stimuli:

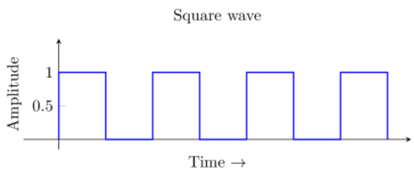
- A single vibration
- A vibration pattern

In the first case, the user can define the duration and the amplitude (a value between 0 and 255) of the vibration. In the second case, the user can create a stimulation pattern, specifying a list of blocks with a given duration and amplitude.

For example, a 4 cycles square wave (8 blocks: 4 with max amplitude, 4 with amplitude zero) can be defined as following:

duration: 500,500,500,500,500,500,500,500

Ampl. : 255,0,255,0,255,0,255,0



Device Validation

Each stimulus is generated with a certain delay with respect to the onset given into the software code. This delay depends on how each modality (audio, visual, tactile) is generated by each device's hardware. These delays are stable and once calculated and corrected, allows the App to deliver true simultaneous multi-modal stimuli. Assuming, for example, that audio modality is the fastest one, and visual one is the slowest, simultaneity is achieved by first delivering the slowest visual stimulus and few milliseconds later the tactile and then the audio one. Validating a new device requires the evaluation of these specific modalities' differences, so that they can be applied to each stimulation. The process of validation developed within our lab first consists in delivering "single aligned" stimuli and measuring, with the aid of an oscilloscope and proper sensors, the delay between each sensory modality.

Full device validation consists in calculating one value for each of the eight possible stimulus types:

1. Audio with Tone (ToneGenerator.startTone)
2. Audio with MediaPlayer API (*MediaPlayer.start*)
3. Audio with AutioTrack API (*AudioTrack.play*)
4. Audio with Native API (*PlaybackEngine.deliver*)
5. Tactile in single modality
6. Tactile in sequence modality
7. Visual: ImageView made visible/invisible
8. Visual: different ImageView loaded at once

Presently, users cannot insert into the App the obtained values and apply the correction but are warmly invited to communicate to us the model’s name and these values to let us add them in the source code. To do it autonomously, developers can access the source code and add their own device.

In MainApplication.kt user can add a new entry to the *devicesDelays* property. The String name of the device is given by: Build.MODEL

```
private var devicesDelays:HashMap<String, DelaysAligner> = hashMapOf(
    "Mi A2 Lite"    to DelaysAligner(4L, 40L, 4L, 0L,5L, 0L, 30L, 53L),
    "SM-A405FN"     to DelaysAligner(0L, 165L, 4L, 0L, 28L, 0L, 20L, 0L),
    "UNKNOWN"       to DelaysAligner(0L, 0L, 0L, 0L, 0L, 0L, 0L, 0L)
)
```